LAB 7: Configuration of Switch, VLAN Configuration and Inter-VLAN Routing

OBJECTIVES

- To configure and manage multiple VLANs on a network switch.
- To set up trunk links and allocate switch ports to their respective VLANs.
- To implement inter-VLAN communication using a router and verify the routing between VLANs.

THEORY

1. Switch Configuration

A network switch is a device that operates at the Data Link Layer (Layer 2) of the OSI model and is responsible for forwarding data between devices in a network. Unlike hubs, which send data to all connected devices, a switch forwards frames only to the intended destination port. It maintains a MAC address table that maps device addresses to specific ports. This intelligent forwarding reduces unnecessary traffic, minimizes collisions, and improves network performance. Switch configuration also involves setting port modes and organizing traffic through logical grouping.

2. Virtual Local Area Network (VLAN)

A Virtual Local Area Network (VLAN) is a logical network segmentation technique used to divide a physical network into multiple isolated broadcast domains. In a traditional switch network, all connected devices share the same broadcast domain, causing unnecessary traffic. VLANs solve this by assigning devices to different logical groups identified by VLAN IDs. Each VLAN operates as an independent network, preventing broadcasts from reaching other VLANs. This improves network efficiency, enhances security, and allows better management of network resources without changing the physical cabling.

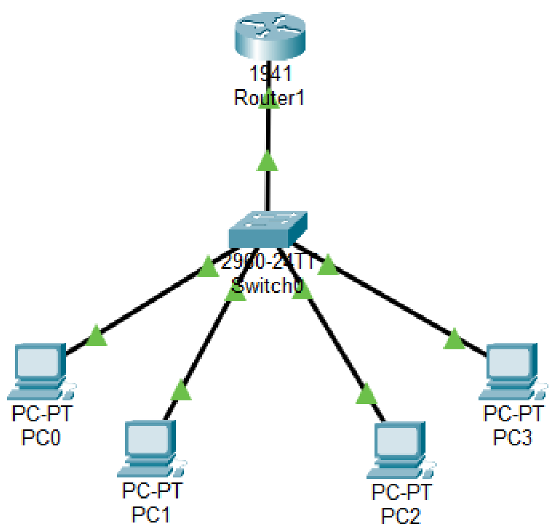
3. VLAN Configuration

Setting up VLANs on a switch involves two main steps: first, defining each VLAN with a unique ID and a descriptive name in the switch's VLAN database; second, placing the appropriate physical ports into those VLANs. Ports designated as access ports carry traffic for a single VLAN only and are typically connected to end devices such as PCs or printers. Ports that need to carry traffic for multiple VLANs simultaneously -for example, an uplink to a router -are instead configured as trunk ports, which tag frames with their respective VLAN IDs using the IEEE 802.1Q standard.

4. Inter-VLAN Routing

Since each VLAN functions as a separate broadcast domain, communication between different VLANs requires a Layer 3 device. One common method is the Router-on-a-Stick configuration, where a single trunk link connects a switch to a router. The router interface is divided into multiple sub-interfaces; each assigned an IP address acting as the default gateway for a VLAN. Using 802.1Q encapsulation, the router identifies VLAN tags and enables communication between different VLANs through one physical connection.

NETWORK TOPOLOGY



CONFIGURATION

VLAN and IP Addressing Scheme

VLAN	Name	Network	Gateway
10	Computer	192.168.10.0/24	192.168.10.1
20	Electronics	192.168.20.0/24	192.168.20.1

Device	IPv4 Address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3	192.168.20.3	255.255.255.0	192.168.20.1

Switch Configuration Commands

1. **Create VLANs**

```
Switch(config)# vlan 10
Switch(config-vlan)# name Computer
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 20
Switch(config-vlan)#name Electronics
Switch(config-vlan) # exit
```

2. **Assigning Ports to VLANs**

```
Switch(config)# interface range fa0/1-2
Switch(config)# switchport mode access
Switch(config) # switchport access vlan 10
Switch(config)# exit
```

```
Switch(config)# interface range fa0/3-4
Switch(config)# switchport mode access
Switch(config)# switchport access vlan 20
Switch(config)# exit
```

3. **Configure Trunk Port (Switch-Router)**

```
Switch(config)# interface fa0/5
Switch(config)# switchport mode trunk
Switch(config)# exit
```

Verification Switch

```
Switch# show vlan brief
```

Router Configuration – Inter-VLAN Routing

```
Router(config)# interface GigabitEthernet 0/0.10
Router(config)# encapsulation dot1Q 10
Router(config)# ip address 192.168.10.1 255.255.255.0
Router(config)# exit
```

Router(config)# interface GigabitEthernet 0/0.20

Router(config)# encapsulation dot1Q 20

Router(config)# ip address 192.168.20.1 255.255.255.0

Router(config)# exit

OBSERVATIONS

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   computer                         active    Fa0/1, Fa0/2
20   Electronics                      active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Ping within Same VLAN (VLAN 10)

PC0 (192.168.10.2) successfully pinged PC1 (192.168.10.3) within the same VLAN. The ping results showed:

- 4 packets sent, 4 packets received -0% packet loss
- Round-trip time: Minimum = 0ms, Maximum = 0ms, Average = 0ms

This confirms that intra-VLAN communication is functioning correctly.

```
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping across Different VLANs (VLAN 10 to VLAN 20)

PC1 (192.168.10.3) successfully pinged PC2 (192.168.20.3) across VLANs through the router. The ping results showed:

- 4 packets sent, 4 packets received -0% packet loss (after initial timeout)
- Round-trip time: Minimum = 0ms, Maximum = 0ms, Average = 0ms

The first packet timed out due to ARP resolution, which is expected behavior. Subsequent packets succeeded, confirming that Inter-VLAN routing via Router-on-a-Stick is working correctly.

DISCUSSION AND CONCLUSION

In this lab, we successfully configured VLANs on a Cisco 2960 switch and implemented Inter-VLAN Routing using the Router-on-a-Stick method on a Cisco 1941 router. Two VLANs were created: VLAN 10 (Computer) and VLAN 20 (Electronics), with switch ports assigned accordingly. A trunk link was configured between the switch and router to carry traffic for both VLANs.

The router's GigabitEthernet interface was divided into sub-interfaces with 802.1Q encapsulation to enable inter-VLAN communication. Ping tests confirmed that devices within the same VLAN communicate directly, while devices in different VLANs communicate through the router. The lab demonstrated the importance of VLANs in network segmentation and the role of routing in enabling cross-VLAN communication.